

Research Notes on Gradient boosting

Gradient boosting

>> Section 1

If we considered the continuous case, i.e., where H is the set of arbitrary differentiable functions on $\mathbb{R}$, we would update the model in accordance with the following equations

>> Section 2

for $m \geq 1$, where $h_m \in H$ is a base learner function. Unfortunately, choosing the best function h_m at each step for an arbitrary loss function L is a computationally infeasible optimization problem in general. Therefore, we restrict our approach to a simplified version of the problem. The idea is to apply a steepest descent step to this minimization problem (functional gradient descent). The basic idea is to find a local minimum of the loss function by iterating on $F_{m-1}(x)$. In fact, the local maximum-descent direction of the loss function is the negative gradient. Hence, moving a small amount γ such that the linear approximation remains valid:

>> Section 3

where $\gamma > 0$. For small γ , this implies that $L(y_i, F_m(x_i)) \leq L(y_i, F_{m-1}(x_i))$.

>> Section 4

The gradient boosting method assumes a real-valued y . It seeks an approximation $\hat{F}(x)$ in the form of a weighted sum of M functions $h_m(x)$ from some class H , called base (or weak) learners:

>> Section 5

Informal introduction (This section follows the exposition by Cheng Li.) Like other boosting methods, gradient boosting combines weak "learners" into a single strong learner iteratively. It is easiest to explain in the least-squares regression setting, where the goal is to teach a model F to predict values of the form $\hat{y} = F(x)$ by minimizing the mean squared error $\frac{1}{n} \sum_i (\hat{y}_i - y_i)^2$, where i indexes over some training set of size n of actual values of the output variable y :

>> Section 6

Gradient boosting is a machine learning technique based on boosting in a functional space, where the target is pseudo-residuals instead of residuals as in traditional boosting. It gives a prediction model in the form of an ensemble of weak prediction models, i.e., models that make very few assumptions about the data, which are typically simple decision trees. When a decision tree is the weak learner, the resulting algorithm is called gradient-boosted trees; it usually outperforms random forest. As with other boosting methods, a gradient-boosted trees model is built in stages, but it generalizes the other methods by allowing optimization of an arbitrary differentiable loss function.

>> Section 7

$$F_m(x) = F_{m-1}(x) - \gamma \sum_{i=1}^n \nabla F_{m-1}(y_i, F_{m-1}(x_i))$$

>> Section 8

$$F_m(x) = F_{m-1}(x) + \left(\arg \min_{h_m \in H} \left[\sum_{i=1}^n L(y_i, F_{m-1}(x_i) + h_m(x_i)) \right] \right)(x)$$

>> Section 9

where parameter ν is called the "learning rate". Empirically, it has been found that using small learning rates (such as $\nu < 0.1$) yields dramatic improvements in models' generalization ability over gradient boosting without shrinking ($\nu = 1$). However, it comes at the price of increasing computational time both during training and querying: lower learning rate requires more iterations.

>> Section 10

Tree size The number J of terminal nodes in the trees is a parameter which controls the maximum allowed level of interaction between variables in the model. With $J = 2$ (decision stumps), no interaction between variables is allowed. With $J = 3$ the model may include effects of the interaction between up to two variables, and so on. J can be adjusted for a data set at hand. Hastie et al. comment that typically $4 \leq J \leq 8$ work well for boosting and results are fairly insensitive to the choice of J in this range, $J = 2$ is insufficient for many applications, and $J > 10$ is unlikely to be required.

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>> Section 11

$$F_0(x) = \arg \min_{h_m \in H} \sum_{i=1}^n L(y_i, h_m(x_i)),$$
$$F_0(x) = \underset{\{h_m\} \in \{\mathcal{H}\}}{\arg \min} \sum_{i=1}^n L(y_i, h_m(x_i)).$$

>> Section 12

Usage Gradient boosting can be used in the field of learning to rank. The commercial web search engines Yahoo and Yandex use variants of gradient boosting in their machine-learned ranking engines. Gradient boosting is also utilized in High Energy Physics in data analysis. At the Large Hadron Collider (LHC), variants of gradient boosting Deep Neural Networks (DNN) were successful in reproducing the results of non-machine learning methods of analysis on datasets used to discover the Higgs boson. Gradient boosting decision tree was also applied in earth and geological studies – for example quality evaluation of sandstone reservoir.

>> Section 13

If the algorithm has M stages, at each stage m ($1 \leq m \leq M$), suppose some imperfect model F_m (for low m , this model may simply predict $\hat{y}_i$ to be $\bar{y}$, the mean of y). In order to improve F_m , our algorithm should add some new estimator, $h_m(x)$. Thus,